

Canonical heights for abelian group actions of maximal dynamical rank

(Joint work with Fei Hu)

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1. Dynamical degree, Height function & Arithmetic degree

- Let X be a smooth projective variety of dimension n over $\overline{\mathbb{Q}}$.
- Denote by $\text{Aut}(X)$ the group of all automorphisms of X .
- The p -th dynamical degree $d_p(g)$ of $g \in \text{Aut}(X)$ is defined as

$$d_p(g) := \max\{|\lambda| : \lambda \text{ is an eigenvalue of } g^*|_{H^{p,p}(X, \mathbb{C})}\},$$

i.e., the spectral radius of the induced linear map $g^*|_{H^{p,p}(X, \mathbb{C})}$, or equivalently, taking an ample divisor H , we have

$$d_p(g) = \lim_{k \rightarrow \infty} \left(\int_X (g^k)^*(H^p) \cdot H^{n-p} \right)^{1/k}.$$

- The maximum of logarithm of dynamical degrees is the algebraic entropy of g , i.e.,

$$h_a(g) := \max_{0 \leq p \leq n} \log d_p(g).$$

1. Dynamical degree, Height function & Arithmetic degree

Theorem (Gromov and Yomdin)

The algebraic entropy coincides with the topological entropy $h_t(g)$, a dynamical invariant measuring the divergence of the orbits.

Log concavity of dynamical degrees (Khovanskii and Teissier)

$\exists u \leq v$ s.t. $1 = d_0(g) < \cdots < d_u(g) = \cdots = d_v(g) > \cdots > d_n(g) = 1$.

Theorem (Dinh-Sibony, 2004/2005; cf. Truong, Dang)

The dynamical degrees are bi-meromorphic invariants.

Definition (Positive entropy)

An element $g \in \text{Aut}(X)$ is of **positive entropy**, if $h_t(g) > 0$, or equivalently $d_1(g) > 1$. A group $G \subseteq \text{Aut}(X)$ is of positive entropy, if every non-trivial element in G is of positive entropy.

1. Dynamical degree, Height function & Arithmetic degree

Theorem (Weil's Height Machine)

There exists a unique homomorphism

$$h_X: \text{Pic}(X)_{\mathbb{R}} \rightarrow \{\text{functions } X(\overline{\mathbb{Q}}) \rightarrow \mathbb{R}\} / \{\text{bounded functions } X(\overline{\mathbb{Q}}) \rightarrow \mathbb{R}\}$$

satisfying the following properties.

- *Let D be a very ample divisor on X and $\phi_D: X \hookrightarrow \mathbb{P}^N$ the associated embedding. Then we have*

$$h_{X,D} = h \circ \phi_D + O(1),$$

where h is the absolute logarithmic height on $\mathbb{P}^N$.

- *Let $\pi: X \rightarrow Y$ be a morphism of smooth projective varieties and $D_Y \in \text{Pic}(Y)_{\mathbb{R}}$. Then we have*

$$h_{X,\pi^*D_Y} = h_{Y,D_Y} \circ \pi + O(1).$$

1. Dynamical degree, Height function & Arithmetic degree

Theorem (Weil's Height Machine (continued))

- Let D_1, D_2 be $\mathbb{R}$ -divisors on X . Then we have

$$h_{X,D_1+D_2} = h_{X,D_1} + h_{X,D_2} + O(1).$$

- Let D be an effective integral divisor on X . Then $h_{X,D} \geq O(1)$ outside the base locus $\text{Bs}(D)$ of D .
- Let $H, D \in \text{Pic}(X)_{\mathbb{R}}$ be $\mathbb{R}$ -divisors with H ample and D algebraically equivalent to zero. Then there is a constant $C > 0$ such that

$$h_{X,D} \leq C \sqrt{h_{X,H}^+},$$

where $h_{X,H}^+ := \max(1, h_{X,H})$.

The terms $O(1)$ only depend on varieties, divisors, and morphisms, but are independent of rational points of varieties.

1. Dynamical degree, Height function & Arithmetic degree

Definition (Arithmetic degrees)

Let h_H be a Weil height function associated with an ample divisor H on X . Set $h_H^+ = \max(1, h_H)$. For each $x \in X(\overline{\mathbb{Q}})$, we define the **arithmetic degree** of f at x by

$$\alpha_f(x) := \lim_{m \rightarrow \infty} h_H^+(f^m(x))^{1/m} \in \mathbb{R}_{\geq 1}.$$

From (Kawaguchi-Silverman, 2016(a) and (b)), the limit always exists and is independent of the choice of the ample divisor H .

Conjecture (Kwaguchi-Silverman Conjecture)

Let $f: X \rightarrow X$ be a surjective endomorphism. Then for any point $x \in X(\overline{\mathbb{Q}})$, if the forward f -orbit $\mathcal{O}_f(x) := \{f^m(x) : m \in \mathbb{Z}_{\geq 0}\}$ of x is Zariski dense in X , then $\alpha_f(x) = d_1(f)$.

1. Dynamical degree, Height function & Arithmetic degree

A few remarks on KSC

- Kawaguchi-Silverman's conjecture asserts the property of arithmetic degrees associated with a dominant rational map. Here, we only consider a rather special situation, i.e., automorphisms.
- KSC has been successfully proved in many cases, leveraging canonical height theory. The application of our main result follows this strategy. Roughly speaking, one aims to construct a height function associated with a nef and big divisor D such that $h_D \circ g^m = d_1(g^m)h_D$.
- An alternative geometric approach to address KSC involves the Equivariant Minimal Model Program, as developed by Meng-Zhang.
- For a comprehensive overview of the current state of EMMP and KSC, we refer to two recent survey papers:
- (Meng-Zhang, 2023) <https://arxiv.org/abs/2311.16369v1>.
- (Matsuzawa, 2023) <https://arxiv.org/abs/2311.15489v1>.

2. Group actions of maximal dynamical rank

Motivated by Smale's work, Dinh and Sibony studied the structure for an abelian subgroup of $\text{Aut}(X)$.

Theorem (Dinh-Sibony, 2004)

Let $G \subseteq \text{Aut}(X)$ be an abelian subgroup where X is a compact Kähler manifold of $\dim n \geq 2$. Then

- The subset $U \subseteq G$ of the elements of zero entropy is a subgroup.
- There is a free abelian subgroup $H \subseteq G$ which is of positive entropy and has rank $\leq n - 1$ such that $G \cong U \times H$.
- If $\text{rank } H = n - 1$ (being maximal), the group U is finite.

Example: complex torus; (Dinh-Sibony, 2004)

The rank estimates in the above theorem are optimal. As mentioned in (Dinh-Sibony, 2004), for each n , there is an n -dim complex torus T such that $\text{Aut}(T)$ contains $G \cong \mathbb{Z}^{\oplus n-1}$, of positive entropy.

2. Group actions of maximal dynamical rank

Zhang proved a theorem of Tits type for compact Kähler manifolds; then, Dinh-Sibony's rank estimates were extended to the solvable group case.

Theorem (Zhang, 2009; cf. Campana-Wang-Zhang, 2014)

Let $G \subseteq \operatorname{Aut}(X)$ be a subgroup where X is a compact Kähler manifold of $\dim n \geq 2$. Then either G contains a free non-abelian subgroup $\mathbb{Z} * \mathbb{Z}$, or there is a finite-index solvable subgroup H such that the subset N of zero entropy elements of H is normal in H and H/N is free abelian of rank $\leq n - 1$. In particular, $\operatorname{Aut}(X)$ satisfies Tits alternative.

- Tits' theorem: every linear group G over $\mathbb{R}$ satisfies Tits alternative, i.e., any subgroup of G either has a free non-abelian subgroup, or is virtually solvable (possesses a finite-index solvable subgroup).
- Tits alternative for $\operatorname{Aut}(X)$ was first conjectured in the joint work (Keum-Oguiso-Zhang, 2009).
- (Hu, 2020) extends Zhang's Tits alternative to $\operatorname{char} > 0$.

2. Group actions of maximal dynamical rank

Question (Dinh-Sibony, 2004; Dinh-Hu-Zhang, 2015)

Classify n -dim compact Kähler manifolds or projective varieties admitting a free abelian group of automorphisms of rank r ($r \leq n - 1$) which is of positive entropy.

- If X is a smooth complex compact surface admitting a positive entropy automorphism, then X is either a complex torus, an Enriques surface, a K3 surface or a rational surface (Cantat, 1999).
- In higher dimensions, if X is a normal projective variety of $\dim n \geq 3$ admitting the action of the group $G := \mathbb{Z}^{\oplus n-1} \subseteq \text{Aut}(X)$ of positive entropy, then either X is rationally connected, or X is birational to a quotient of a complex torus (Zhang, 2016).
- Zhang's result can be extended to klt compact Kähler threefolds (Z., 2022/2023).

3. Main results and applications

Theorem (Hu-Z., 2023; cf. (Kawaguchi, 2008) for the surface case)

Let X be a smooth projective variety of dimension n over $\overline{\mathbb{Q}}$. Let $G \cong \mathbb{Z}^{n-1}$ be a free abelian group of automorphisms of X such that every nontrivial element of G has positive entropy. Then there exist a function $\hat{h}_G$ on $X(\overline{\mathbb{Q}})$ and a G -invariant Zariski closed proper subset Z of X such that:

- $\hat{h}_G$ is a Weil height function with respect to a nef and big $\mathbb{R}$ -divisor D ;
- $\hat{h}_G$ satisfies the Northcott property on $(X \setminus Z)(\overline{\mathbb{Q}})$, i.e., for any $d \in \mathbb{Z}_{>0}$ and $N \in \mathbb{R}$, the set

$$\{x \in (X \setminus Z)(\overline{\mathbb{Q}}) : [\mathbb{Q}(x) : \mathbb{Q}] \leq d, \hat{h}_G(x) \leq N\}$$

is finite, where $Z := \mathbb{B}_+(D)$ is the augmented base locus of D ;

- For any $x \in (X \setminus Z)(\overline{\mathbb{Q}})$, one has $\hat{h}_G(x) \geq 0$; further, $\hat{h}_G(x) = 0$ if and only if x is g -periodic for any $g \in G$.

3. Main results and applications

What is the nef and big $\mathbb{R}$ -divisor D in our main result?

Theorem (Hu-Z., 2023; cf. (Dinh-Sibony, 2004))

Let X be a smooth projective variety of dimension n defined over $\overline{\mathbb{Q}}$ and $G \cong \mathbb{Z}^{n-1}$ a free abelian group of automorphisms of X of positive entropy. Then the following assertions hold.

- There are exactly n distinct multiplicative group characters $\chi_i : G \rightarrow (\mathbb{R}_{>0}, \times)$ with $1 \leq i \leq n$, where each χ_i is associated with a common nef eigendivisor D_i , i.e., $g^* D_i \equiv \chi_i(g) D_i$ for any $g \in G$.
- $D_1 \cdots D_n \in \mathbb{R}_{>0}$; in particular, $D := \sum_{i=1}^n D_i$ is a nef and big $\mathbb{R}$ -divisor on X .
- For any $1 \leq i \leq n$, there is some $g_i \in G$ such that $\chi_j(g_i) < 1$ for all $j \neq i$, $\chi_i(g_i) = d_1(g_i)$, and $g_i^* D_i \sim_{\mathbb{R}} d_1(g_i) D_i$.

Caution: The automorphisms $g_1, \dots, g_n$ may not be the generators of G . Instead, any $n - 1$ of them generate a finite index subgroup of G .

3. Main results and applications

We give a few remarks on the above construction and its usage in the proof of our main result.

- The construction of the common nef eigendivisors D_i is essentially due to (Dinh-Sibony, 2004), who initially constructed $n - 1$ of them and then separately constructed the last one. This separateness however, prevents a direct use of their construction to define a canonical height in an appropriate way.
- Our work herein is to prove that there exist n distinguished automorphisms $g_1, \dots, g_n$ in the rank $n - 1$ abelian group G such that $\chi_i(g_i) = d_1(g_i)$ and $\chi_j(g_i) < 1$ for all $j \neq i$. Consequently, all n common nef eigendivisors D_i share the same status.
- By (Call-Silverman, 1993), for each pair (g_i, D_i) as described above, there exists a unique nef canonical height function $\hat{h}_{D_i, g_i}$. As the sum $\sum_i D_i$ forms a nef and big $\mathbb{R}$ -divisor, the corresponding sum $\sum_i \hat{h}_{D_i, g_i}$ of nef canonical height functions turns out to be a Weil height function (denoted by $\hat{h}_G$) satisfying the Northcott property.

3. Main results and applications

Corollary (Existence of Zariski dense G -orbits and potential density)

Under the assumption of maximal dynamical rank, there exists a rational point $x \in X(\overline{\mathbb{Q}})$ such that the G -orbit $\mathcal{O}_G(x) := \{g(x) : g \in G\}$ of x is Zariski dense in X . Moreover, if we assume that $X := Y \times_{\operatorname{Spec} K} \operatorname{Spec} \overline{\mathbb{Q}}$ for some Y defined over a number field K , then there exists a finite field extension L/K such that $Y_L(L)$ is Zariski dense in Y_L .

Caution: The above result does NOT imply for any $g \in G$, there is a Zariski dense g -orbit.

Corollary (KSC)

Under the assumption of maximal dynamical rank, for any $g \in G$ and any point $x \in (X \setminus \mathbb{B}_+(D))(\overline{\mathbb{Q}})$, we have

$$\alpha_g(x) = \begin{cases} 1 & \text{if } x \text{ is } g\text{-periodic,} \\ d_1(g) & \text{if } x \text{ is not } g\text{-periodic.} \end{cases}$$

3. Main results and applications

Motivated from the surface case due to (Kawaguchi, 2008) and (Kawaguchi-Silverman, 2016), we obtain the following application.

Corollary (Bounded height property and height counting function)

Under the assumption of maximal dynamical rank, the following assertions hold for any $\text{id} \neq g \in G$.

- 1 The subset below

$$S := \{x \in (X \setminus Z)(\overline{\mathbb{Q}}) : x \text{ is } g\text{-periodic}\}$$

is of bounded height, i.e., for a fixed ample divisor H_X , there is a constant C such that $h_{H_X}(s) \leq C$ for all $s \in S$.

- 2 For any $x \in (X \setminus Z)(\overline{\mathbb{Q}})$ which is not g -periodic, one has

$$\lim_{T \rightarrow +\infty} \frac{\#\{m \in \mathbb{Z}_{\geq 0} : h_{H_X}(g^m(x)) \leq T\}}{\log T} = \frac{1}{\log d_1(g)}.$$

3. Main results and applications

Remark (Doable extensions of our main results to other cases)

- All of our main results and applications herein can be extended to normal projective varieties over $\overline{\mathbb{Q}}$ without any difficulties. The two modifications are (1) the Hodge index theorem in the singular setting and (2) Weil's height theory for $\mathbb{R}$ -Cartier divisors.
- It is also possible to extend our main results to positive characteristics as the language in this line has also been developed due to Truong, Dang, and Hu.
- Note that the Hodge index theorem in the singular case (or even in the positive characteristic) is well-developed due to Nakayama, Zhang, Hu, Meng-Z., etc.

4. Open questions

Question 1

Under the assumption of maximal dynamical rank, for each non-trivial element $g \in G$, does there exist a Zariski dense orbit on X ?

(Amerik, 2011) verifies that there exists a g -forward orbit which is infinite. Our corollary shows that there exists a G -forward Zariski dense orbit. However, it remains unclear whether there exists a Zariski dense g -orbit for a single element $g \in G$.

Question 2

Can we verify or disprove the existence of a rationally connected smooth projective of dimension ≥ 3 which is not birational to an étale quotient of an abelian variety but satisfies the maximal dynamical assumption?

Using (Zhang, 2016), we can verify KSC under the further assumption that X is not rationally connected. However, the equivariant minimal model program for the case of rationally connected varieties is still unknown.

4. Open questions

Question 3

Under the assumption of maximal dynamical rank, following (Silverman, 1991), we can also define a function

$$\hat{H}_G: X(\overline{\mathbb{Q}}) \rightarrow \mathbb{R} \quad \text{by} \quad \hat{H}_G(x) = \prod_{i=1}^n \hat{h}_{D_i, g_i}(x).$$

Due to the projection formula, for any $g \in G$, $\prod_{i=1}^n \chi_i(g) = 1$. So

$$\hat{H}_G \circ g = \prod_{i=1}^n \hat{h}_{D_i, g_i} \circ g = \prod_{i=1}^n \chi_i(g) \hat{h}_{D_i, g_i} = \prod_{i=1}^n \hat{h}_{D_i, g_i} = \hat{H}_G.$$

In other words, the function $\hat{H}_G$ is G -invariant. Moreover, $\hat{H}_G(x) = 0$ if and only if x is g -periodic for any $g \in G$. Let $x \in (X \setminus \mathbb{B}_+(D))(\overline{\mathbb{Q}})$ be arbitrary. Can we bound $\hat{H}_G(x)$ in terms of $\hat{h}_G(x)$?

Thank you for your attention!